命名空间和接合点
ONTAP 9

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目录

命名空间和接合点 1
命名空间和接合点概述 1
什么是典型的 NAS 命名空间架构 2

命名空间和接合点

命名空间和接合点概述

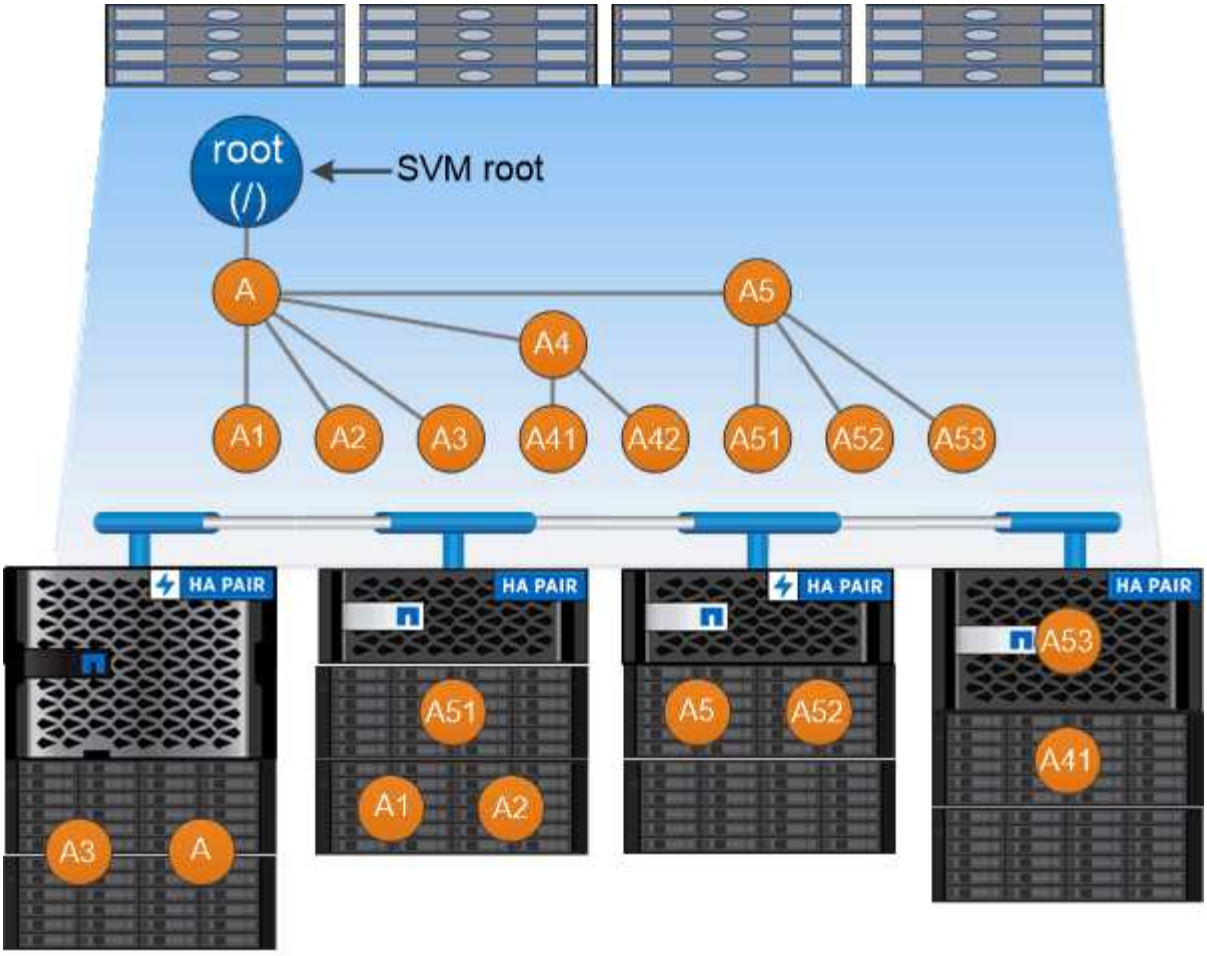
nas_namespaces_ 是指在 *junction points* 处联合在一起的卷的逻辑分组，用于创建单个文件系统层次结构。具有足够权限的客户端可以访问命名空间中的文件，而无需指定文件在存储中的位置。集群中的任何位置都可以驻留未分配的卷。

NAS 客户端不会挂载包含相关文件的每个卷，而是挂载 nfs *export* 或访问 SMB _share 。 _ 导出或共享表示整个命名空间或命名空间中的中间位置。客户端仅访问挂载在其访问点下方的卷。

您可以根据需要向命名空间添加卷。您可以直接在父卷接合下方或卷中的目录上创建接合点。名为 "`vol3` " 的卷的卷接合路径可能为 ` /vol1/vol2/vol3` ，或 ` /vol1/dir2/vol3` ，甚至是 ` /dir1/dir2/vol3` 。此路径称为 _junction path…

每个 SVM 都有一个唯一的命名空间。SVM 根卷是命名空间层次结构的入口点。

> 要确保在发生节点中断或故障转移时数据仍然可用，您应为 SVM 根卷创建一个 *load-sharing mirror* 副本。



A namespace is a logical grouping of volumes joined together at junction points to create a single file system hierarchy.

以下示例将在 SVM vs1 上创建一个名为 "`home4` " 的卷，该卷的接合路径为 ` /eng/home` ：

```
cluster1::> volume create -vserver vs1 -volume home4 -aggregate aggr1
-size 1g -junction-path /eng/home
[Job 1642] Job succeeded: Successful
```

什么是典型的 NAS 命名空间架构

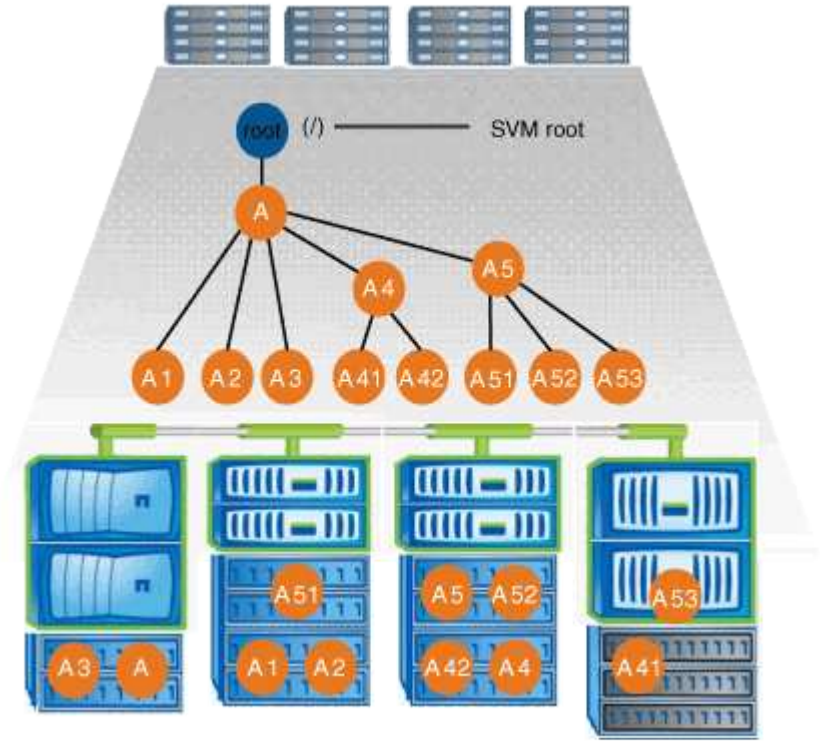
在创建 SVM 名称空间时，您可以使用几种典型的 NAS 命名空间架构。您可以选择符合业务和工作流需求的命名空间架构。

命名空间的顶部始终为根卷，以斜杠（/）表示。根下的命名空间架构分为三个基本类别：

- 一个分支树，与命名空间根只有一个接合点
- 多个分支树，多个接合点指向命名空间的根
- 多个独立卷，每个卷都有一个指向名称空间根的单独接合点

包含单个分支树的命名空间

包含单个分支树的架构在 SVM 命名空间的根上具有一个插入点。单个插入点可以是接合卷，也可以是根下的目录。所有其他卷都挂载在单个插入点（可以是卷或目录）下的接合点处。

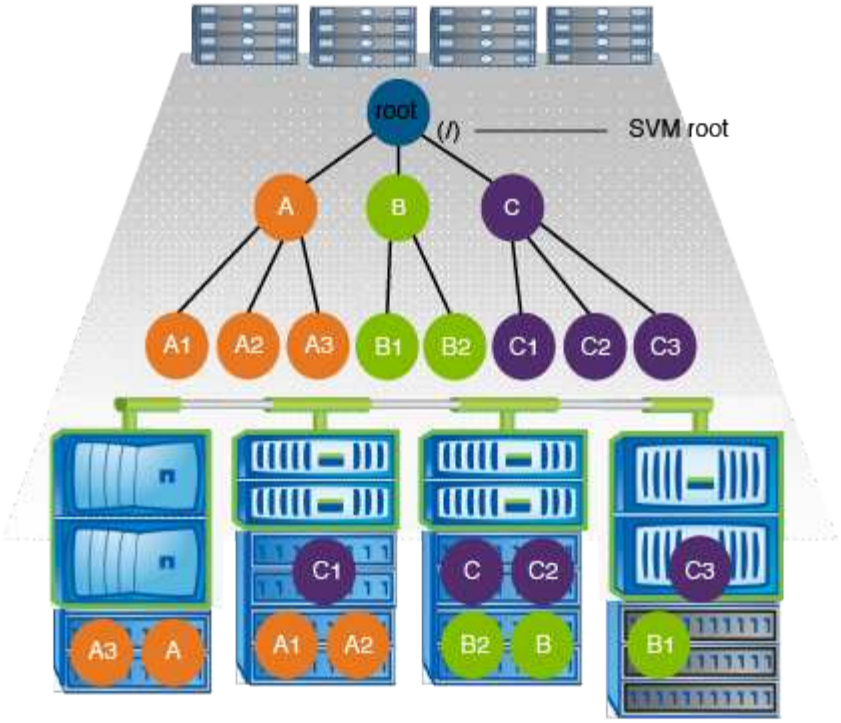


例如，具有上述命名空间架构的典型卷接合配置可能类似于以下配置，其中所有卷都在单个插入点（即名为 `data` 的目录）下接合：

```
                      Junction                          Junction
Vserver Volume        Active   Junction Path            Path Source
------- ------------- -------- --------------------     -----------
vs1     corp1         true     /data/dir1/corp1         RW_volume
vs1     corp2         true     /data/dir1/corp2         RW_volume
vs1     data1         true     /data/data1              RW_volume
vs1     eng1          true     /data/data1/eng1         RW_volume
vs1     eng2          true     /data/data1/eng2         RW_volume
vs1     sales         true     /data/data1/sales        RW_volume
vs1     vol1          true     /data/vol1               RW_volume
vs1     vol2          true     /data/vol2               RW_volume
vs1     vol3          true     /data/vol3               RW_volume
vs1     vs1_root      -        /                        -
```

包含多个分支树的命名空间

包含多个分支树的架构在 SVM 命名空间的根目录中具有多个插入点。插入点可以是接合卷，也可以是根下的目录。所有其他卷都挂载在插入点下方的接合点（可以是卷或目录）。

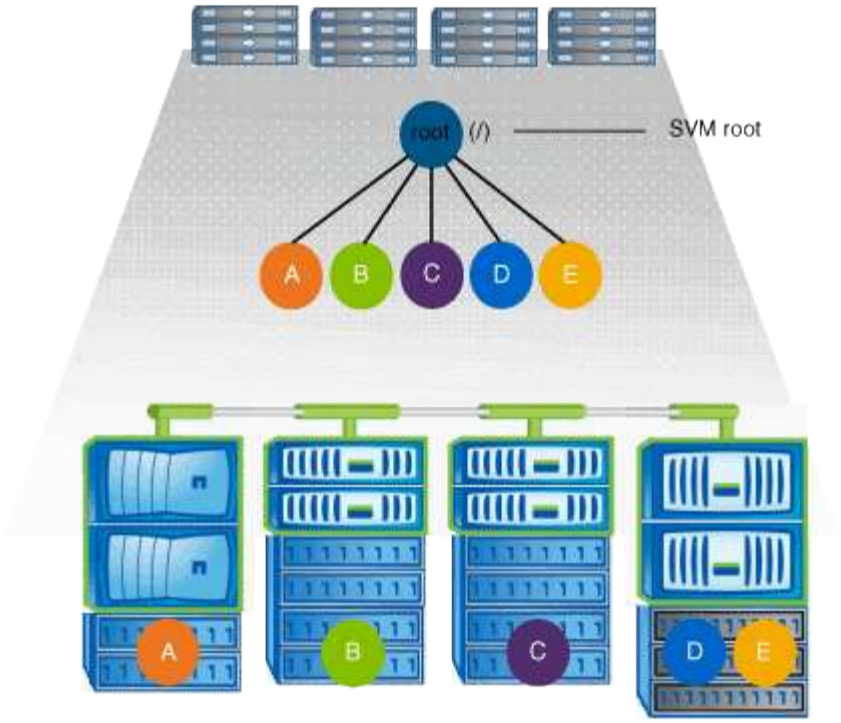


例如，具有上述命名空间架构的典型卷接合配置可能类似于以下配置，其中有三个插入点指向 SVM 的根卷。两个插入点是名为 `data` 和 "`projects` " 的目录。一个插入点是名为 "`audit` " 的接合卷：

```
                      Junction                          Junction
Vserver Volume        Active   Junction Path            Path Source
------- ------------- -------- --------------------     -----------
vs1     audit         true     /audit                   RW_volume
vs1     audit_logs1   true     /audit/logs1             RW_volume
vs1     audit_logs2   true     /audit/logs2             RW_volume
vs1     audit_logs3   true     /audit/logs3             RW_volume
vs1     eng           true     /data/eng                RW_volume
vs1     mktg1         true     /data/mktg1              RW_volume
vs1     mktg2         true     /data/mktg2              RW_volume
vs1     project1      true     /projects/project1       RW_volume
vs1     project2      true     /projects/project2       RW_volume
vs1     vs1_root      -        /                        -
```

包含多个独立卷的命名空间

在具有独立卷的架构中，每个卷都有一个插入点指向 SVM 命名空间的根；但是，卷不会接合到另一个卷下。每个卷都有一个唯一的路径，可以直接在根下接合，也可以在根下的目录下接合。



例如，具有上述命名空间架构的典型卷接合配置可能类似于以下配置，其中有五个插入点指向 SVM 的根卷，每个插入点表示一个卷的路径。

```
                      Junction                          Junction
Vserver Volume        Active   Junction Path            Path Source
------- ------------- -------- --------------------     -----------
vs1     eng           true     /eng                     RW_volume
vs1     mktg          true     /vol/mktg                RW_volume
vs1     project1      true     /project1                RW_volume
vs1     project2      true     /project2                RW_volume
vs1     sales         true     /sales                   RW_volume
vs1     vs1_root      -        /                        -
```

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